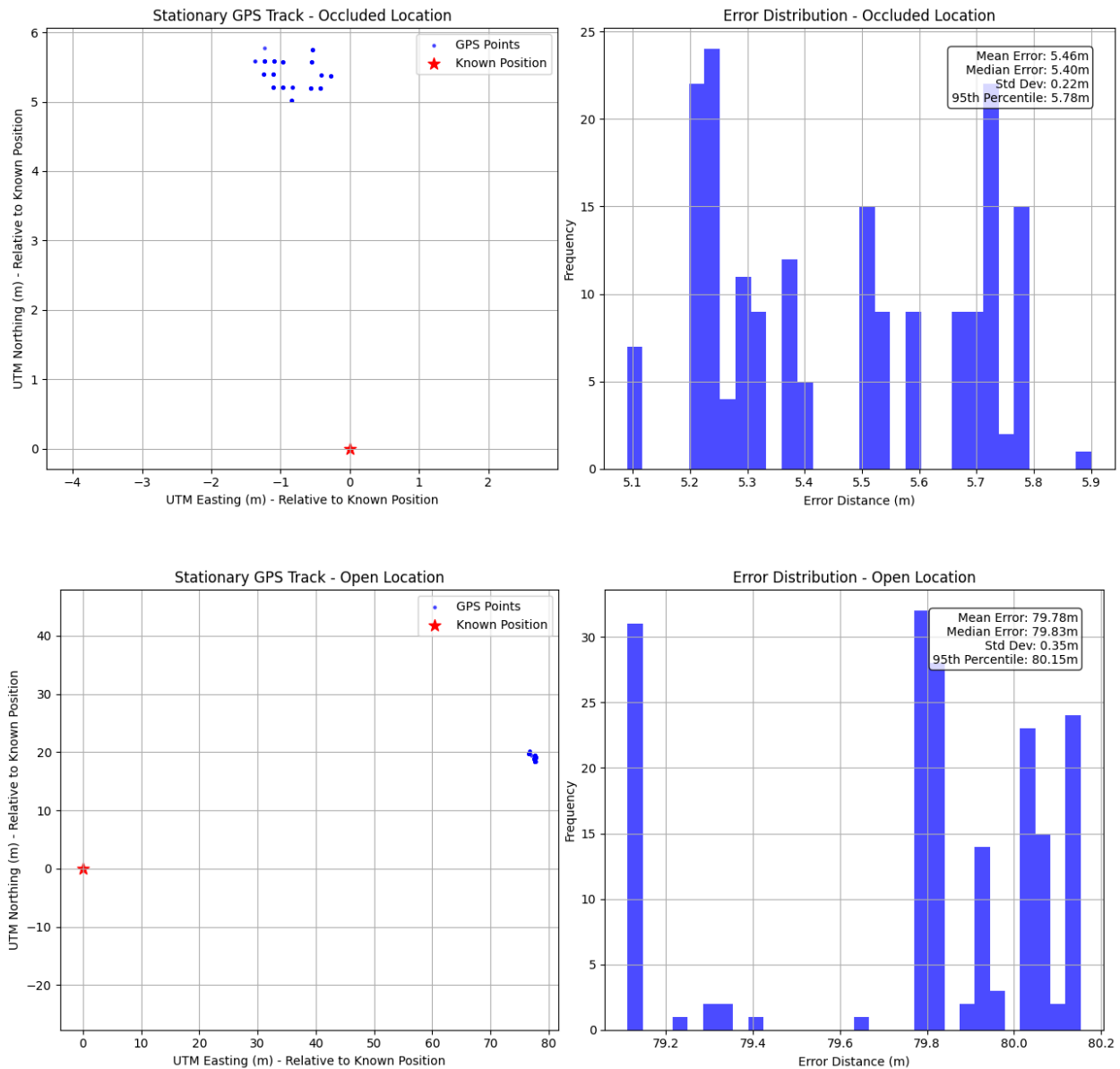


NOTE: Due to Linux environment issues, I was unable to collect data with my own driver. The data used for this analysis is from Sofia Lanzi (lanzi.s@northeastern.edu).

Part 1. Stationary Data

Scatter Plots and Histograms, Northing vs Easting



3.

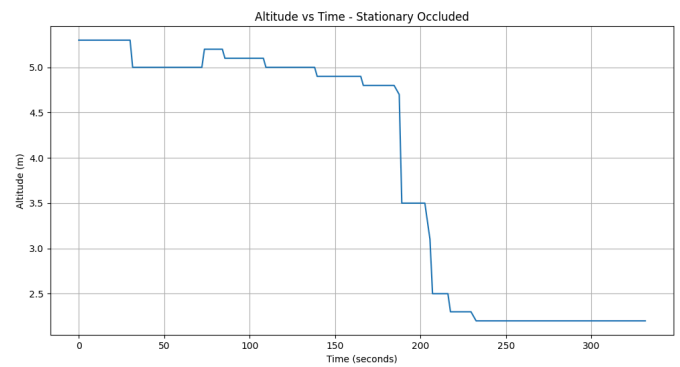
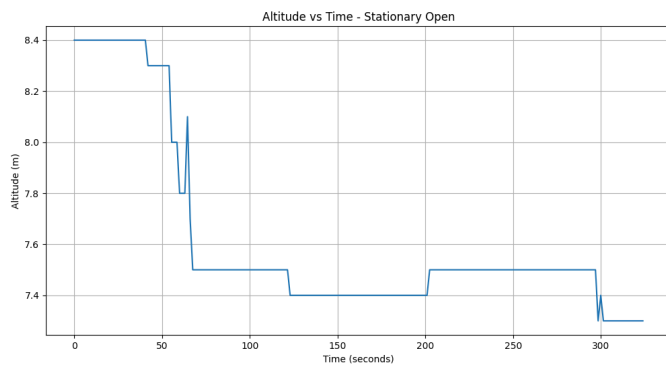
a.) What is your quantitative error estimate from your “known” position?

	Occluded	Open
Mean HDOP	1.57	1.40
Median Error (m)	5.40	79.83

b.) Does this error value make sense given your hdop value and GPS error in general?

Yes, this error makes sense. The HDOP values are in a good range, but not the best, and the value for the occluded data is within the expected range for areas prone to multipath error, as it was recorded standing in between ISEC and EXP on campus. In the case of the open area data, there are a few possible sources for the large error. The most probable explanation would be that the baseline distance between the base station and the user station was initialized incorrectly, which would account for the large difference between the known position and the readings. The standard deviation between readings is low, indicating that the precision of the readings aligned with the expected performance, which makes this a likely reason for the estimated error.

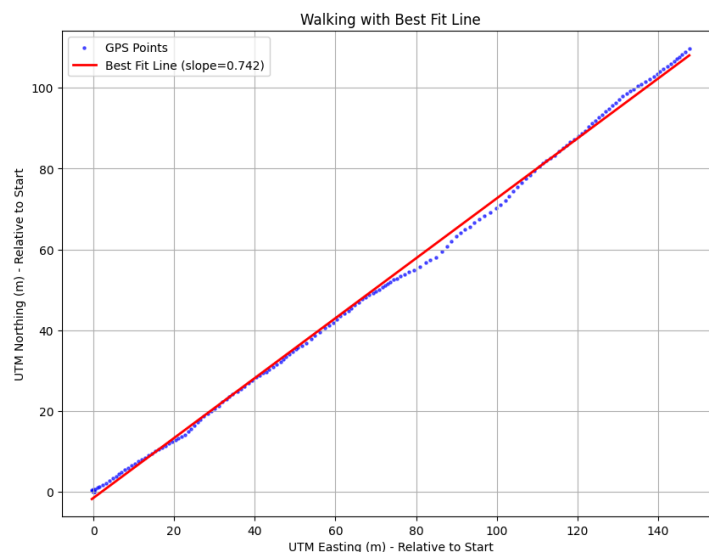
4. Altitude vs Time



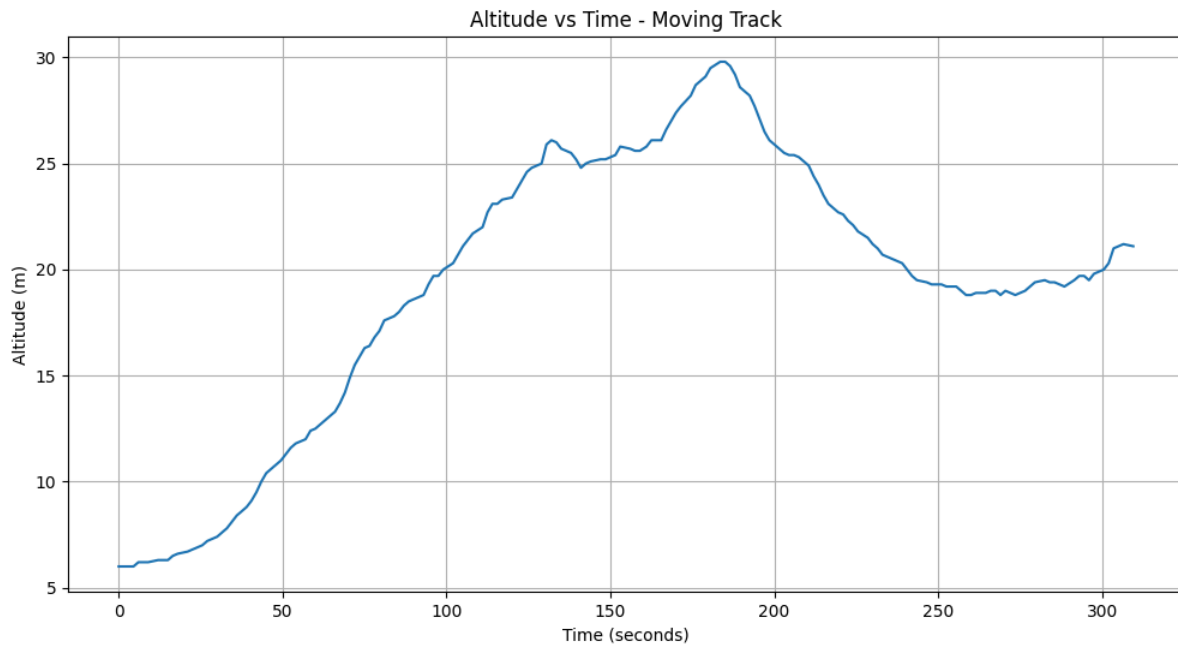
Part 2. Walking Data

Scatter Plot of Northing vs Easting with Fitted Line

a.) RMS error: 1.04 m



b.) Altitude vs Time



Discussion

a.) How do your estimated error values change for stationary vs. moving data?

The estimated error is much lower for the moving data when compared against the stationary data error.

b.) Can you explain why this result is the case? What does this say about GPS navigation when moving for our receiver?

This result makes a lot of sense because of the Kalman filter that is implemented. The model of the Kalman filter relies on linearization around each data point, i.e. $y = ax + b$, which is a reasonable and good representation of our walking data, as it was recorded while walking in a straight line. Since we compute the error of the dataset against a linear regression, the reported error value is low. Additionally, we avoid the correction factor error that skewed our estimated error result in the stationary analysis.

c.) What are the physically likely source(s) of error in these data sets?

As previously mentioned, multipath error and signal quality from the base station are likely sources. Additionally, atmospheric and internal GPS heat effects can also contribute to the total error. It's also possible that, at the time of data collection, the fourth satellite used for time correction was positioned suboptimally in its orbit, affecting trilateration and the quality of time correction.

